

Cognitive Systems: Neural Networks and Applied AI

Exercise 2

Modeling and Simulating Biological Neural Networks

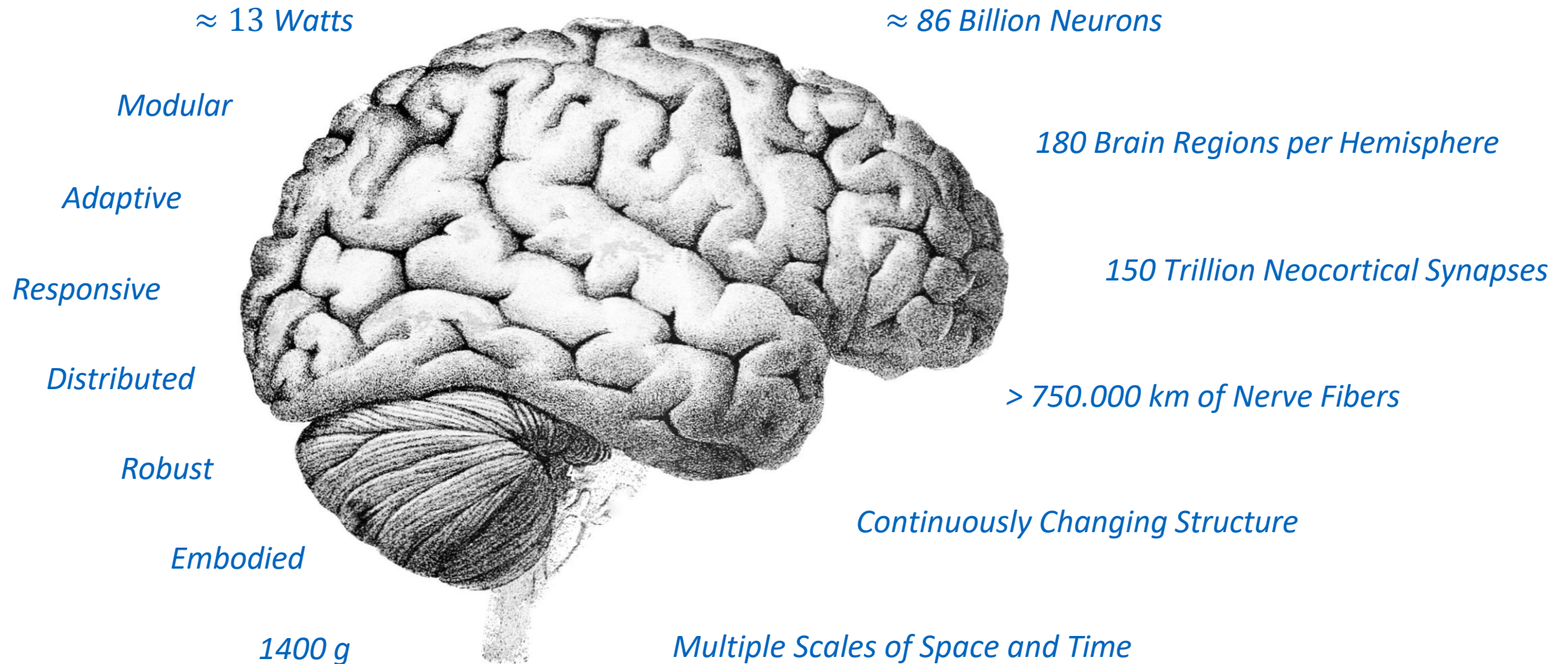
Prof. Dr.-Ing. habil. Alois Knoll

Topics in this Exercise

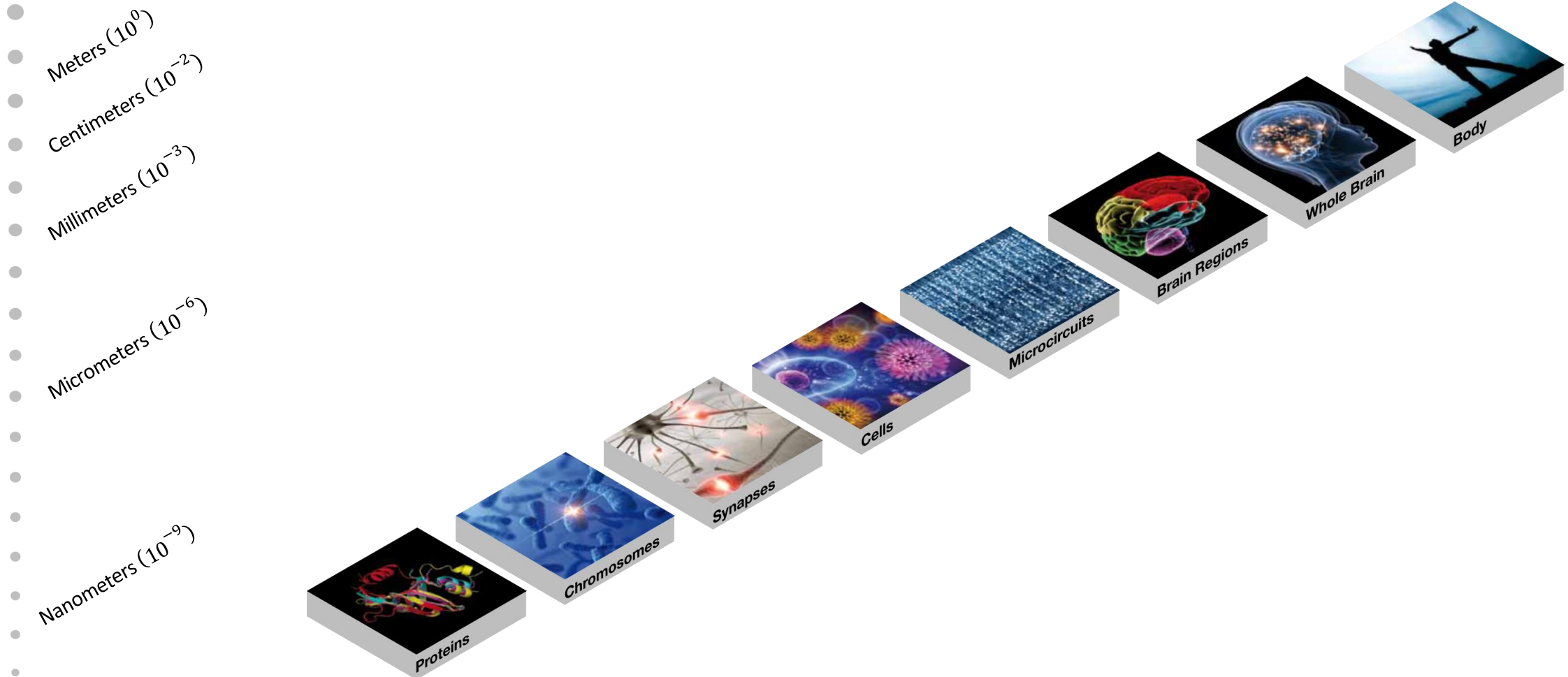
How do biological neurons work?

- What is computational neuroscience?
- What are the building blocks of biological neurons?
- How can biological neurons be modeled mathematically?

Cognition – The State of the Art



Multilevel Organization of the Brain



Adapted from "The Human Brain Project: A Report to the European Commission"

Multilevel Organization of the Brain



Adapted from "The Human Brain Project: A Report to the European Commission"

Neuroscience: The Science of Neural Cognition in the Brain

“The ultimate goal of neural science is to understand how the flow of electrical signals through neural circuits gives rise to mind – to how we perceive, act, think, learn, and remember.”

From Principles of Neural Science (Kandel et al., 2012)

Computational Neuroscience

Definition E5.1: Computational Neuroscience (also: Theoretical Neuroscience)

“Computational neuroscience is the field of study in which mathematical tools and theories are used to investigate brain function. It can also incorporate diverse approaches from electrical engineering, computer science, and physics in order to understand how the nervous system processes information.”

From <https://www.nature.com/subjects/computational-neuroscience>

Biological Neurons

The Main Components of a Neuron

Every neuron is comprised of three main components:

Soma

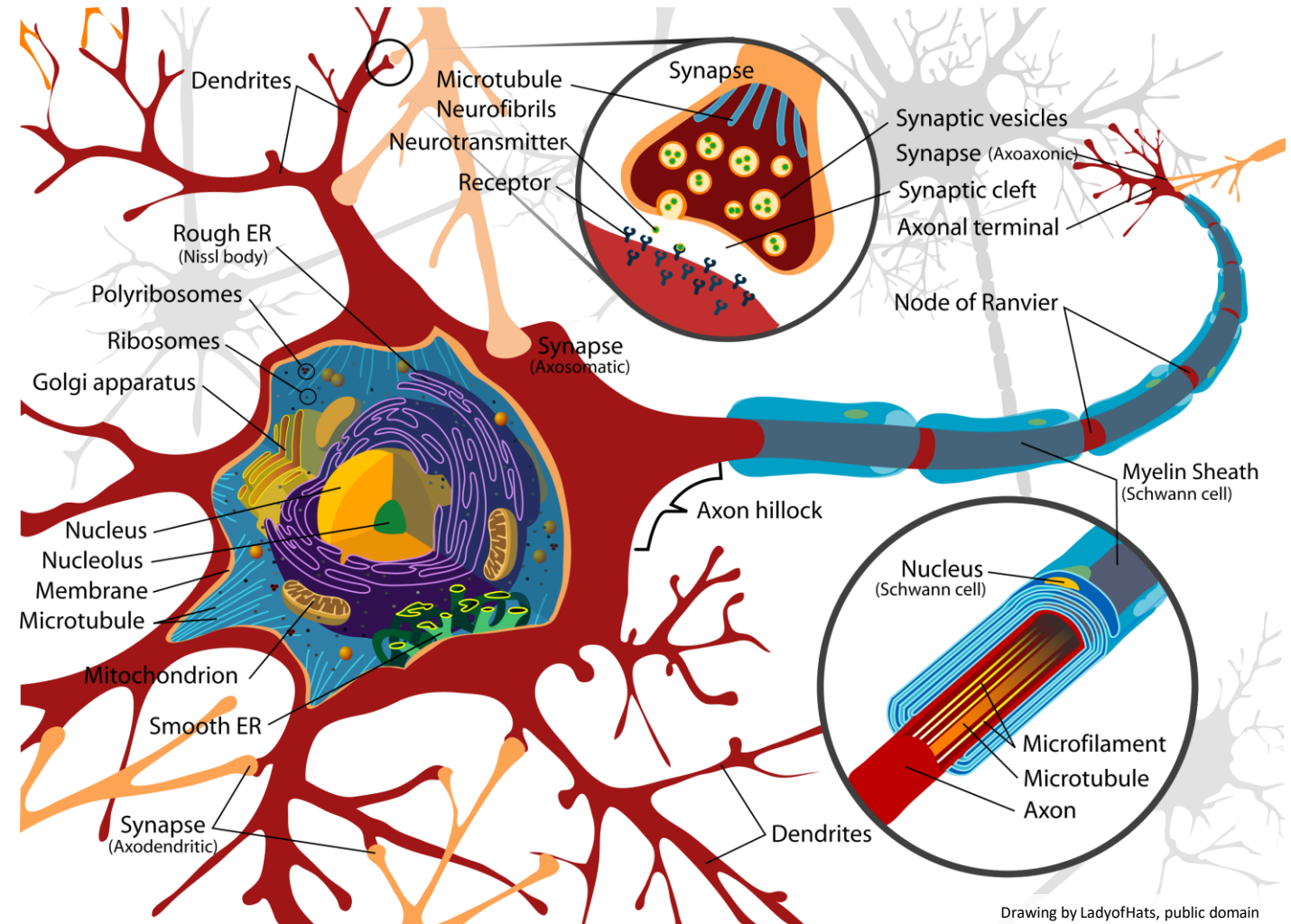
The soma is the cell body containing the nucleus.

Dendrites

The dendrites receive electrical impulses from other neurons through **synapses**.

Axon

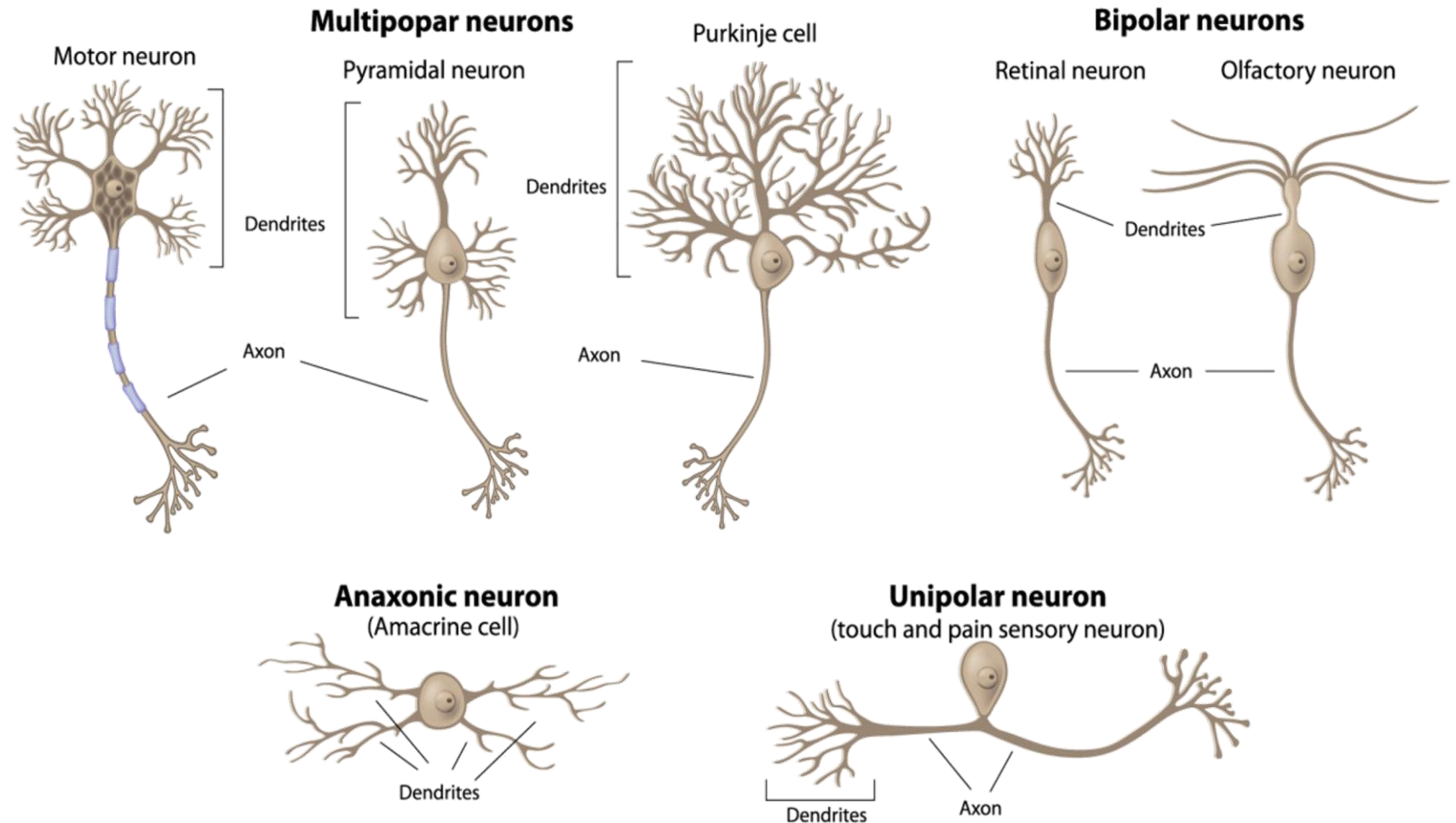
The axon emerges from the soma at the axon hillock and conducts electrical impulses to other neurons.



Neuron Types and Morphologies

The **polarity** of a neuron denotes the number of different processes that originate from the soma.

More generally, neurons are classified by their functionality into **sensory neurons**, **interneurons**, and **motor neurons**.



Multipolar Neurons



From <http://blausen.com/en/video/structure-of-a-multipolar-neuron/>

The Basic Electrophysiology of a Neuron

Information processing and exchange in biological nervous systems are based on **electrical signals**. Unlike electrical circuits, charges are not carried by electrons but by **ions** (Na^+ , K^+ , Cl^-).

- The **cell membrane** of neurons is a bilipid layer that **insulates** the inside of the cell from the outside (the membrane acts as a **capacitor**)
- **Ion channels** are proteins in the membrane that can open and close to allow for a **passive** exchange of ions between the inside and the outside of the neural cell (the membrane has a **resistance**)
 - **Selectivity**: selective ion channels only let a specific type of ion pass
 - **Regulation**: the opening and closing of an ion channel can be gated by the binding of a **ligand**, by **voltage**, or by **mechanical force**

The Resting Membrane Potential

In the idle state of a neuron, the sum of the different ion flows between the inside and the outside of the neuron results in an equilibrium state called the **resting membrane potential**.

- By default, ions cross the channels to decrease the **chemical gradient** (concentration difference) between the inside and the outside of the neuron
- The ion flow implies a flow of charges, which results in a **voltage** between the inside and the outside of the neuron
- The voltage gradient is directed opposite to the charge gradient
- The voltage at the equilibrium point of the chemical gradient and the voltage gradient is the **resting membrane potential** of the neuron

Computing the Resting Membrane Potential

The resting membrane potential of a neuron is mainly determined by the potassium (K^+) concentration since the cell membrane has a very high **permeability** for potassium at the resting state. It is therefore possible to approximate the resting state potential by computing the **Nernst potential** of potassium. In general, the Nernst potential is computed as:

$$E_{Nernst} = \frac{RT}{nF} \ln \frac{[Cation^+]_o}{[Cation^+]_i} \text{ (sign is flipped for anions)}$$

R : gas constant

T : temperature in Kelvin

n : number of charges per ion

F : Faraday constant

Computing the Resting Membrane Potential

Assuming that $T = 293\text{ K}$ and substituting the constants yields the following equation:

$$E_{Nernst} = 58 \log \frac{[\text{Cation}^+]_o}{[\text{Cation}^+]_i} mV$$

Approximation of the Resting Membrane Potential

Outside concentration: $[\text{K}^+]_o = 20mM$

Inside concentration: $[\text{K}^+]_i = 400mM$

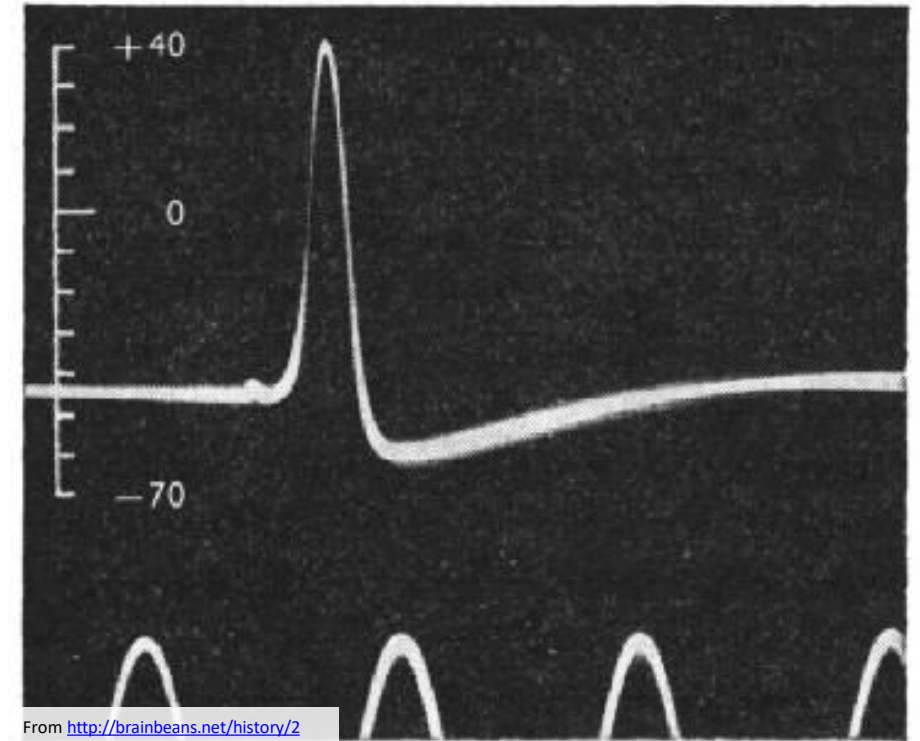
$\rightarrow E_{Nernst} \approx -75mV$

Computing the exact resting potential requires taking into account all ion types ($\rightarrow$ [Goldmann equation](#)).

The Action Potential

Biological neurons exchange information through electrical **impulses** called **action potentials** or **spikes**:

- When a neuron receives spikes from a **presynaptic afferent**, sodium channels open, positive charge flows into the neuron, and the membrane potential increases (**depolarization**)
- As soon as the voltage reaches a threshold, a large number of **voltage-gated sodium channels open**, and the membrane voltage quickly increases to a peak voltage, the **action potential**
- After some delay, **voltage-gated potassium channels** open, and the membrane voltage falls back below the resting state (**hyperpolarization**)

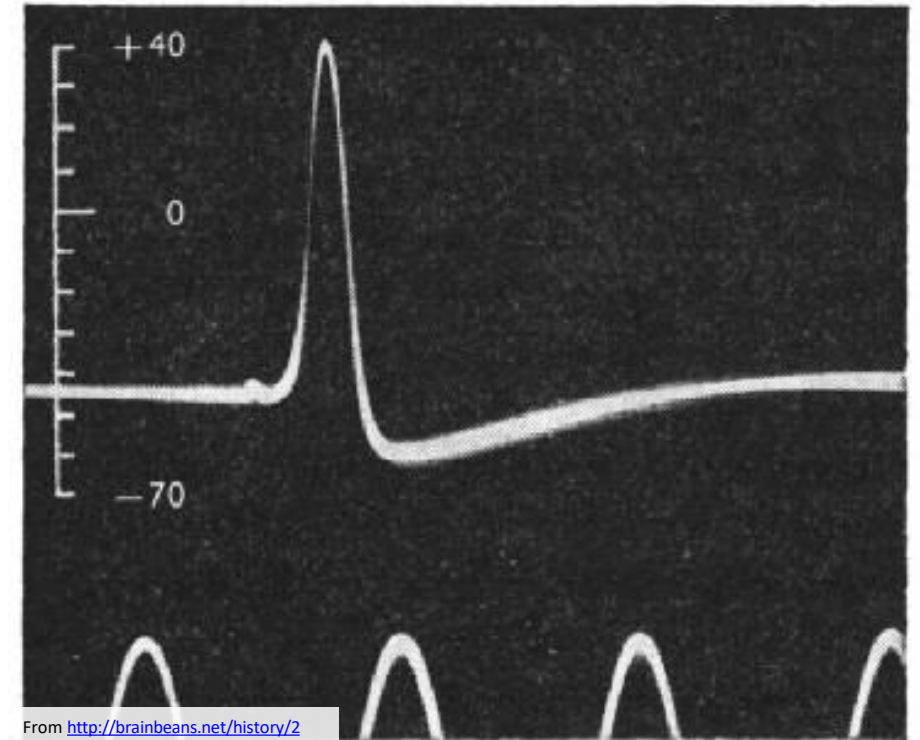


*Early recording of an action potential by Alan **Hodgkin** and Andrew **Huxley** in the **giant axon** of a squid.*

The Action Potential

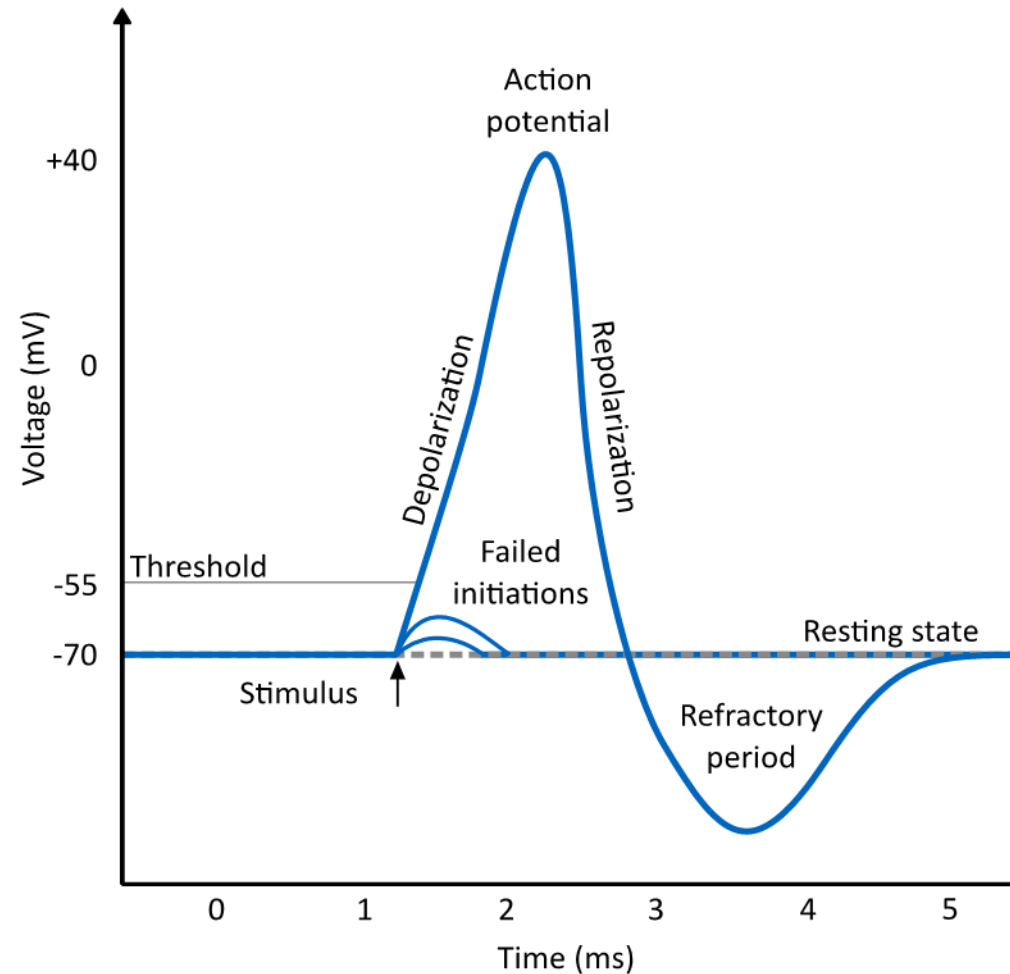
Directly after an action potential, there is a time called the refractory period during which the neuron recovers its resting membrane potential. It is split into two phases:

- **Absolute refractory period**: dead time after a spike during which the neuron cannot emit any new spike (caused by inactivation of voltage-gated sodium channels)
- **Relative refractory period**: eliciting an action potential requires stronger stimuli compared to the resting state of the neuron



*Early recording of an action potential by Alan **Hodgkin** and Andrew **Huxley** in the **giant axon** of a squid.*

Different Phases of an Action Potential



The Action Potential



From <http://blausen.com/en/video/action-potential/>

Action Potential Propagation

The nervous system implements two different modes of propagating action potentials along nerve cells:

Electrotonic Conductance

The passive conduction of charges along the neuron propagates at high speed. However, the strength of the action potential decays with distance. Electrotonic conductance can be improved by increasing the resistance of the membrane and by decreasing the inner resistance of the nerve fiber through a larger diameter.

Active Regeneration of Action Potentials

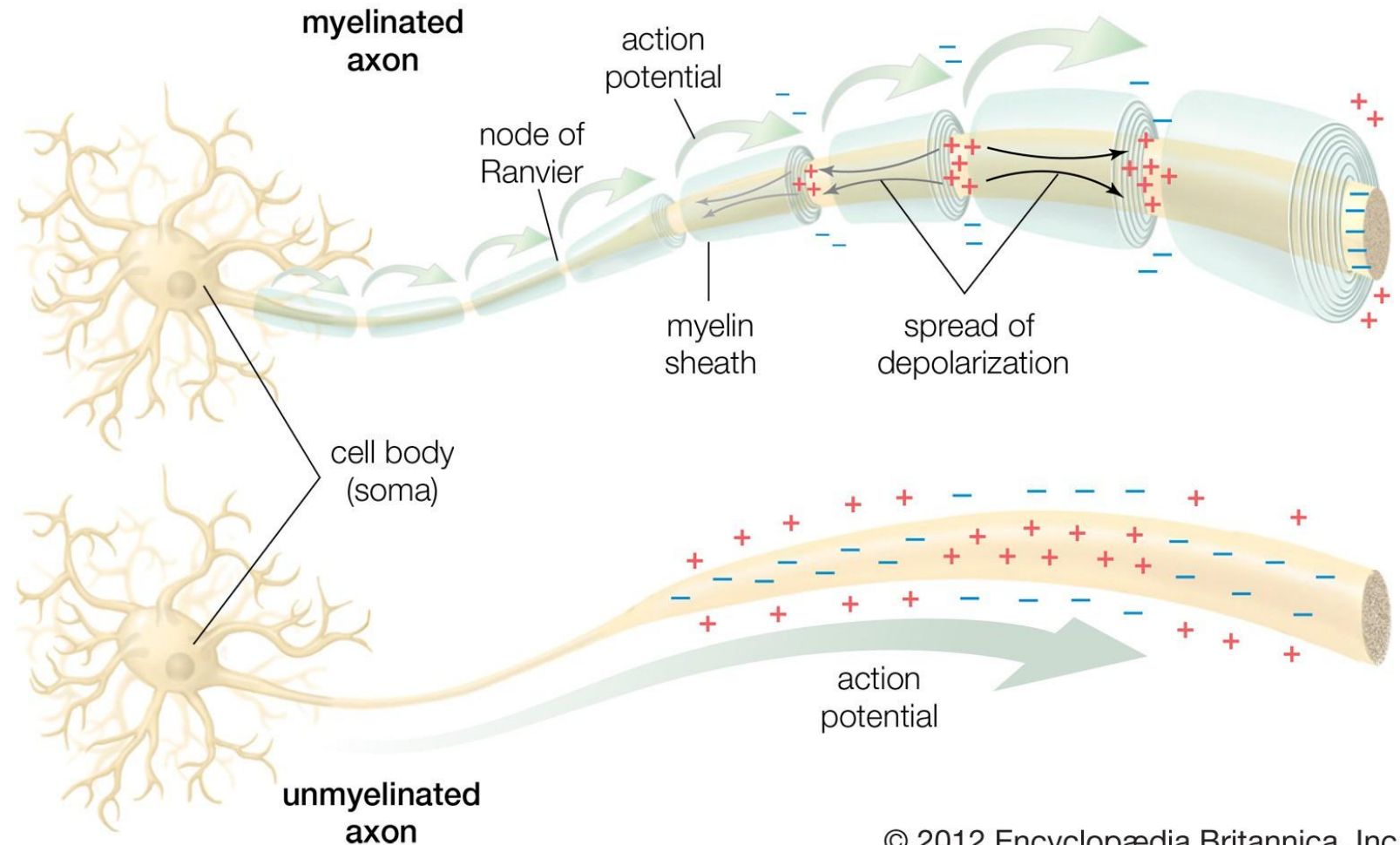
The charges traveling along the neuron cause repeated depolarization in the axon, which results in a regeneration of the action potential. The propagation speed is slow.

Enhancing Neural Signal Transmission

Nerve fibers allow for a fast **saltatory** conduction of action potentials.

In **myelinated** segments of the axon, signals are transmitted through fast electrotonic conduction. The myelin sheath increases the resistance of the membrane.

The action potential is regenerated at gaps in the myelin sheath, the **nodes of Ranvier**.



© 2012 Encyclopædia Britannica, Inc.

The Propagation of Action Potentials



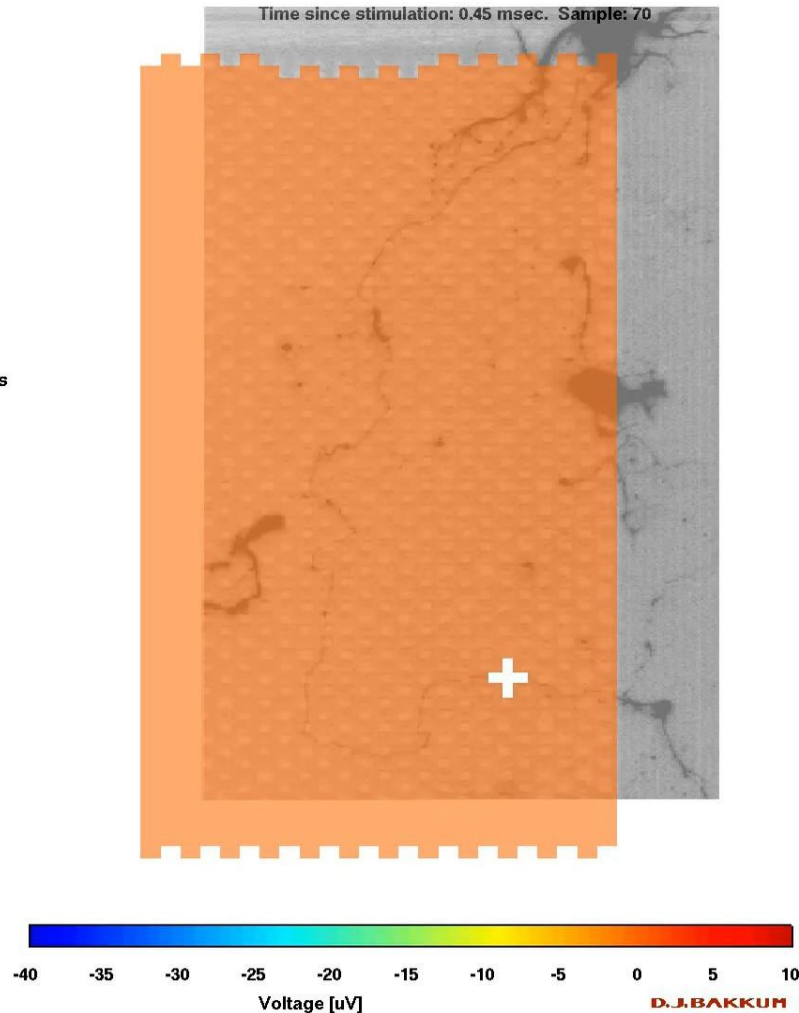
From <http://blausen.com/en/video/action-potential/>

Example: AP Propagation in Cultured Neurons

Supplemental Movie 3 DIV 20

Propagating action potential
in an RFP expressing neuron
evoked by electrical stimulation.

Recording configurations were scanned across
the neuron. The median voltage per electrode
(colored pixels) in response to stimuli
applied to a single electrode (white plus)
60 times per configuration are plotted.
An evoked action potential (yellowish)
travels along an axon toward the soma.
Scale bar: 100 μm .



Summary: Properties of the Action Potential

Action potentials are “digital” events that differ considerably from most other processes in biology:

- **Threshold**: action potentials are elicited when the membrane voltage crosses a specific threshold (typically around -50 mV)
- **All-or-none event**: the shape of the spike does not depend on the history of inputs and is irrelevant
- **Propagation**: spikes are propagated over long distances without decrement through a mechanism of self-regeneration
- **Refractory period**: every spike is followed by a dead time during which the neuron cannot emit any spike; the refractory period, therefore, limits the maximum spike frequency of a neuron

Synapses

Synapses are the sites of **signal transmission** between connected neurons. Every synapse consists of both a **presynaptic** and a **postsynaptic** structure.

Electrical synapses

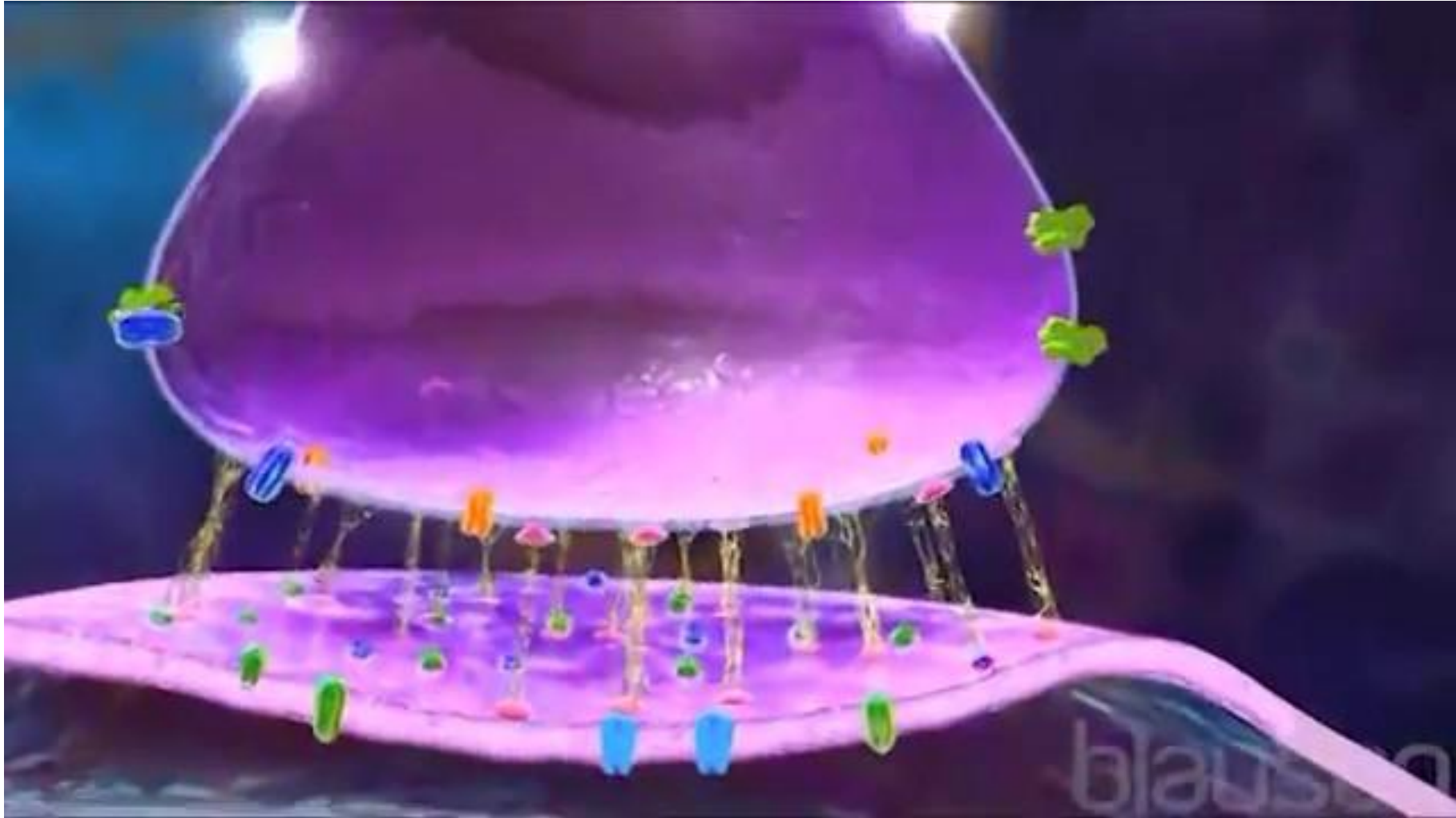
Electrical synapses are based on gap junction channels through which an action potential is directly transferred from the presynaptic to the postsynaptic neuron.

Chemical synapses

In chemical synapses, the synaptic terminals of the presynaptic neuron form a knob-like structure, which is separated from the postsynaptic neuron by a **synaptic cleft**; signals are transferred by a **neurotransmitter** (acetylcholine) that diffuses through the cleft towards receptors of the postsynaptic neuron

- Chemical synapses can be **excitatory** or **inhibitory**
- Synapses can change their strength (→ **learning**)
- Signal transmission can be modulated by **neurotransmitters**

Signal Transmission at a Cholinergic Synapse



From <http://blausen.com/en/video/chemical-synapse-cholinergic-synapse/>

Glial Cells

Most cells in the brain are **glial cells** (grk. glue). They considerably outnumber the neurons – there are about ten times more glia than neurons. Glial cells have a supporting function and do not play a direct role in information processing. There are two main classes of glia:

Microglia

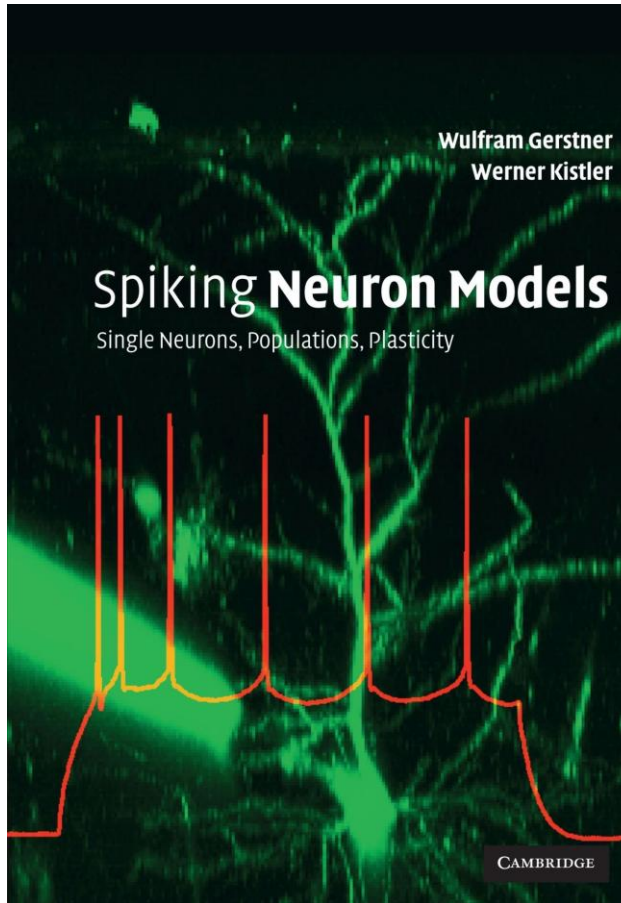
“Microglia are immune system cells that are mobilized to present antigens and become phagocytes during injury, infection, or degenerative diseases.”

Macroglia

Most glial cells are macroglia. The main functions of macroglia include the formation of the myelin sheath around axons and the nutrition of neurons.

Neuron Models and the Foundations of Neural Networks

Key Literature



Free access on <https://lcnwww.epfl.ch/gerstner/PUBLICATIONS/SpikingNeuronM-extracts.pdf>

Central Questions

- How can the complex physiology of biological neurons be modeled and simulated?
- Which level of detail needs to be modeled?
- Is the morphology of a neuron relevant for its computational function?
- How can spikes encode information?

Levels of Modeling

The dynamics of biological neurons are governed by a complex interplay between the **neuron morphology** (geometry of the neuron) and **electrophysiological processes**. Different classes of mathematical models capture these aspects at varying levels of detail:

Detailed Compartmental Models

Detailed compartmental models capture the detailed neuron morphology based on anatomical reconstructions. The continuous geometrical structure of the neuron is discretized into compartments.

Reduced Compartmental Models

Reduced compartmental models are comprised of only a few dendritic compartments to decrease computational complexity and to increase analytical tractability.

Levels of Modeling

Single-Compartment Models (Point Neuron Models)

Single-compartment models completely ignore the morphology of the neuron. Spiking point neuron models capture the flow of ion currents and spike generation.

→ Point neuron models are the most prominent class of neuron models; our focus is therefore on models of this type

Cascade Models

Cascade models are purely functional models that only represent abstract computations, but no biological details.

Black-Box Models

Black-box models do not model any computational function at all, but consider the neuron as an input/output system with defined statistical behavior.

The McCulloch-Pitts Model

The McCulloch-Pitts neuron model was conceived by McCulloch and Pitts in 1943 and is therefore one of the first computational neuron models. Inspired by the all-or-none property of biological neurons, the model is based on a **digital threshold logic**:

- A neuron can have both **excitatory synapses** e_k and **inhibitory synapses** i_l
- If the neuron receives input from any inhibitory synapse i_l , the output of the neuron is 0
- If no inhibitory synapse is active and the sum of active excitatory synapses crosses a threshold, the output of the neuron is 1

The neuron model was originally defined as the basis for a **logical calculus in neural networks**.

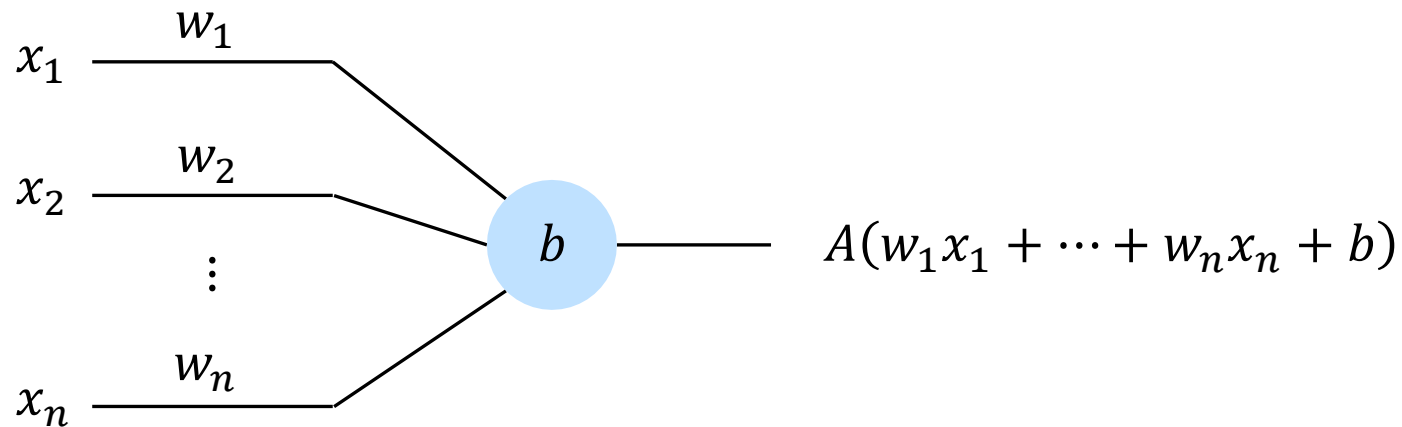
Example: Neurons and Mathematical Operations

Computation	Biophysical Mechanism	Model Level
Addition, Subtraction	Dendritic summation of excitatory and/or inhibitory inputs	Detailed and reduced compartmental models
Subtraction	Shunting mechanism plus integrate-and-fire mechanism	Detailed and reduced compartmental models
Multiplication, Division	Synaptic interaction, gain modulation via synaptic background noise	Detailed and reduced compartmental models
High-pass filter	Firing rate adaptation	Single compartment models
Low-pass filter	Passive membrane properties	Detailed, reduced, and single compartment models
Toggle Switch	Bistable spike generation	Single compartment models

From Herz et al. (2006)

General Analog Neuron Models

The simple McCulloch-Pitts model has been generalized by adding **synaptic weights** and an **activation function**:

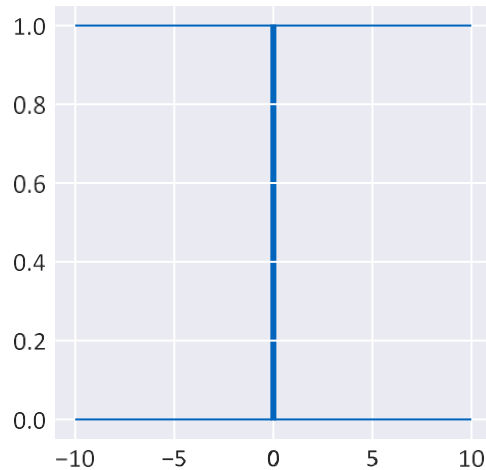


The output $n(x)$ of a neuron n with activation function A , synaptic weights w , bias b , and input x is computed as follows:

$$n(x) = A(w_1x_1 + \dots + w_nx_n + b) = A(w^T x + b)$$

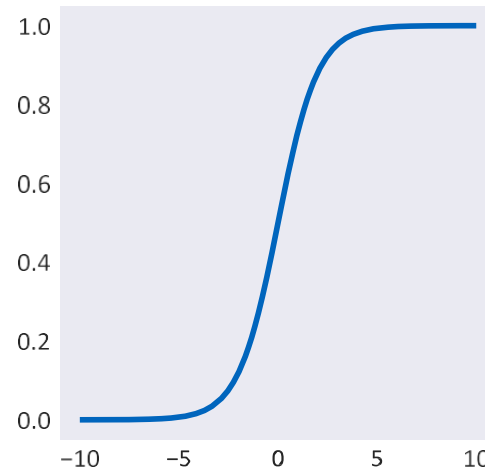
→ **Analog neuron models are generalized linear models!**

Common Activation Functions



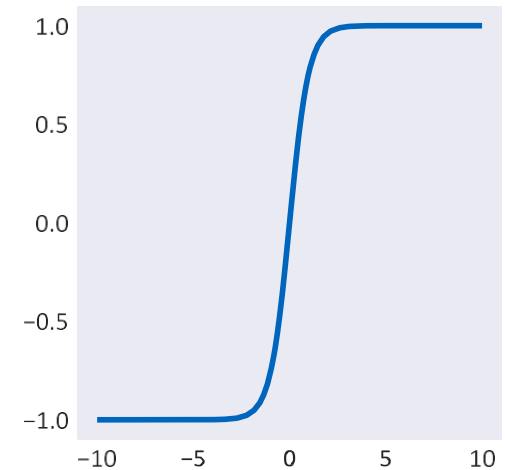
Binary Step Function

$$A(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$



Logistic Function

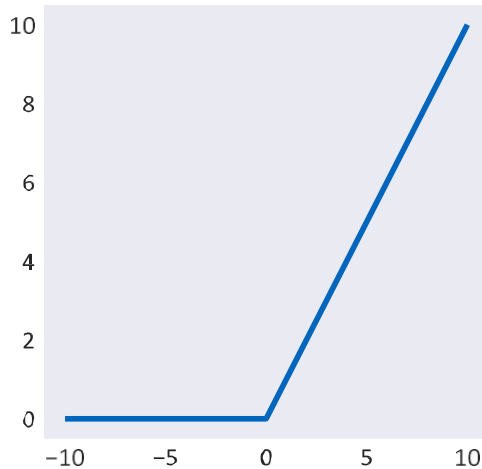
$$A(x) = \frac{1}{1 + e^{-x}}$$



tanh

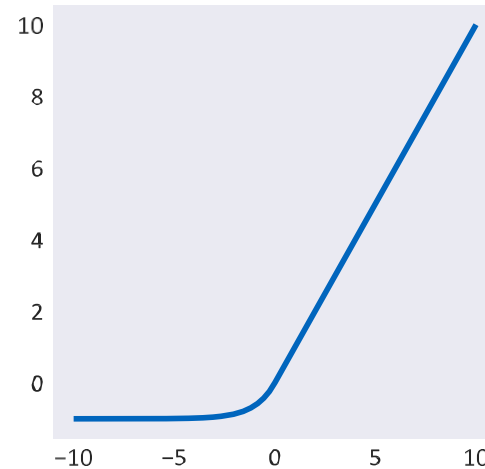
$$A(x) = \tanh x$$

Common Activation Functions



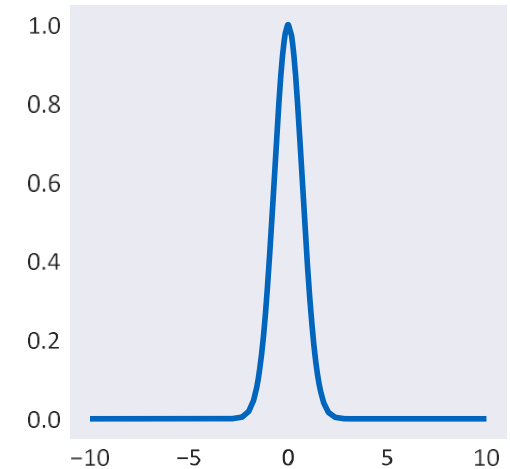
Rectified Linear Unit (ReLU)

$$A(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$



Exponential Linear Unit (ELU)

$$A(x) = \begin{cases} \alpha(e^x - 1), & x < 0 \\ x, & x \geq 0 \end{cases}$$



Gaussian

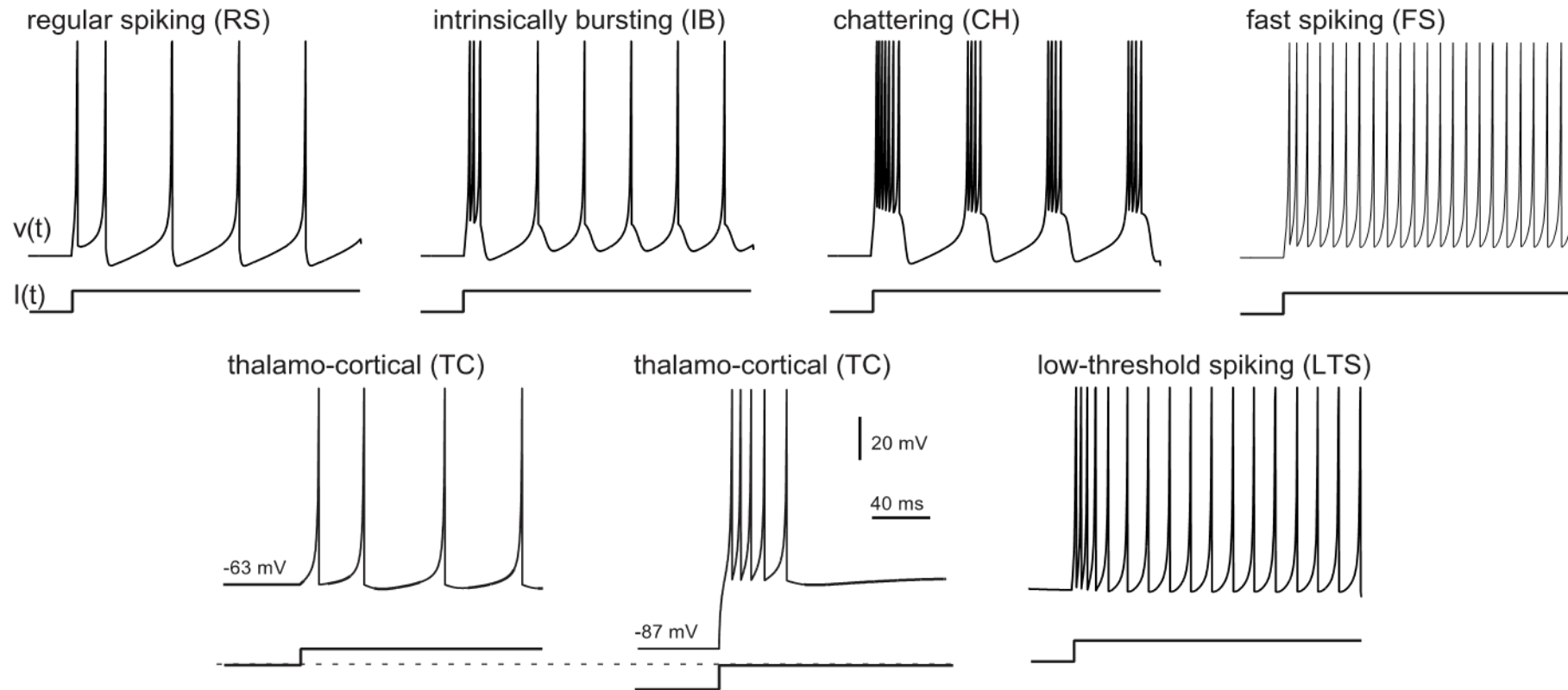
$$A(x) = e^{-\alpha(x-t)^2}$$

Spiking Neuron Models

Analog neuron models are **functions** that are not able to reproduce the **temporal dynamics** of biological neurons. In computational neuroscience, the electrophysiological properties of biological neurons are modeled using **functionally equivalent electrical circuits**:

- The cell membrane is represented by a **capacitor**
 - Flows of different types of ions correspond to **electrical currents**
 - The opening and closing of ion channels corresponds to the change of **conductance** values (inverse of resistance)
- Biological neurons can be emulated using analog electrical circuits, which is a key prerequisite for the design of **analog neuromorphic chips**

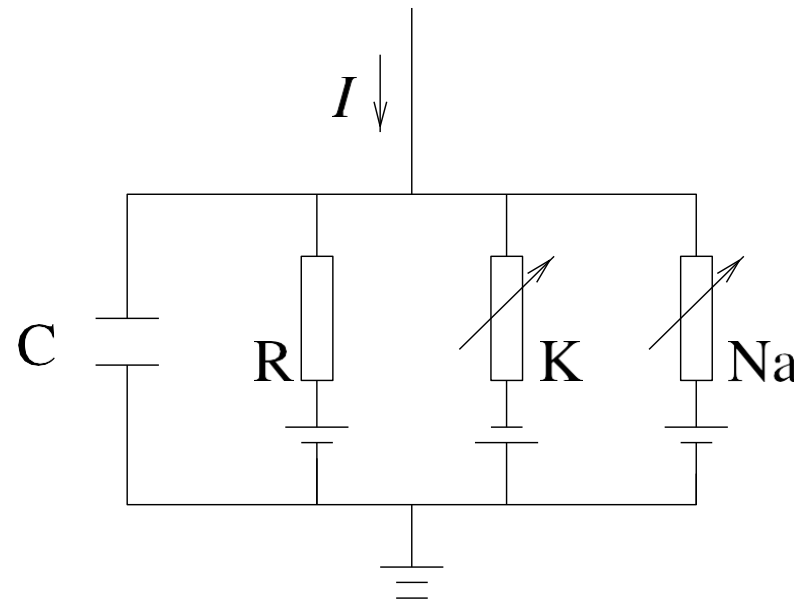
Different Types of Spiking Neurons



Different types of neurons can be characterized by their **voltage response** to a **step current**.

The Hodgkin-Huxley Model

The Hodgkin-Huxley model was derived based on measurements of action potentials in the giant axon of the squid in 1952. The model contains a **single compartment** and is the basis for **conductance-based** neuron models. The overall dynamic behavior is described by the following electrical circuit:



From W. Gerstner and W. M. Kistler, Spiking neuron models: Single neurons, populations, plasticity. Cambridge U.K., New York: Cambridge University Press, 2002.

The Hodgkin-Huxley Model

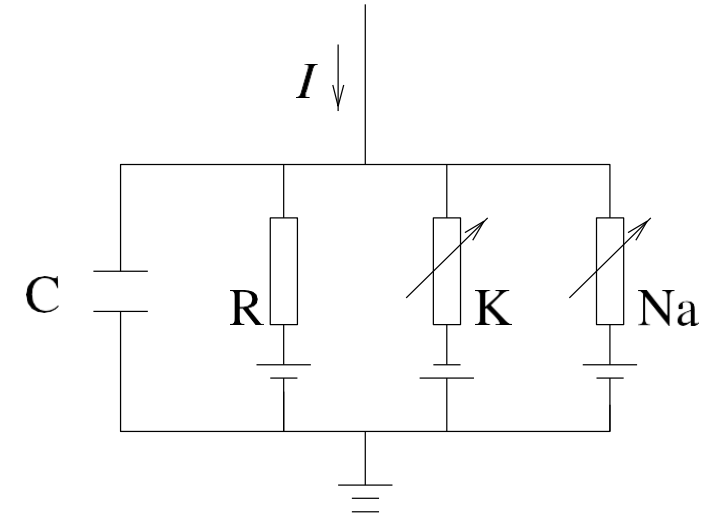
The **passive electrical properties** of the cell membrane are represented by a capacity C and a resistance R . The Nernst potential is emulated by a battery. A current I injected into the cell splits into a capacitive component I_{cap} , which charges the capacitor and three components I_k , which pass through the ion channels (sodium, potassium, leakage):

$$I(t) = I_{cap}(t) + \sum_k I_k(t)$$

Substituting $I_{cap}(t) = C \frac{du}{dt}$ yields:

$$C \frac{du}{dt} = -\sum_k I_k(t) + I(t)$$

u is the **membrane voltage** of the neuron.



From W. Gerstner and W. M. Kistler, Spiking neuron models: Single neurons, populations, plasticity. Cambridge U.K., New York: Cambridge University Press, 2002.

The Hodgkin-Huxley Model

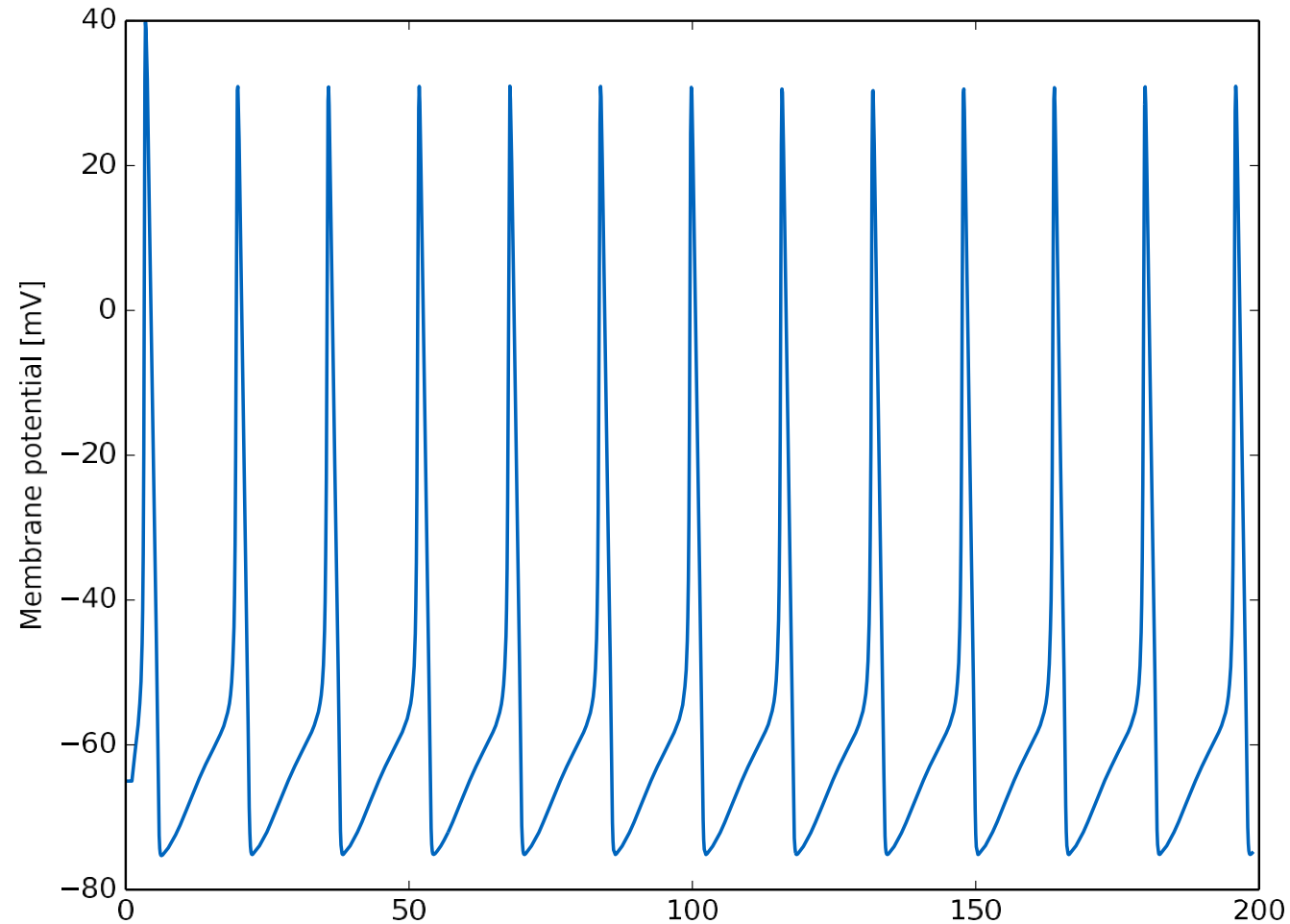
A major drawback of the Hodgkin-Huxley model is its computational complexity, which is mainly caused by the computation of the ionic currents:

$$\sum_k I_k(t) = g_{Na} m^3 h (u - E_{Na}) + g_K n^4 (u - E_K) + g_L (u - E_L)$$

g_{Na} , g_K , and g_L are the conductances of the individual ion channels, and E_{Na} , E_K , and E_L are their **reversal potentials (Nernst potentials)**. m , n , and h are gating variables that model the time-dependent conductances of the ion channels based on another set of differential equations:

$$\begin{aligned} \dot{m} &= \alpha_m(u)(1 - m) - \beta_m(u)m \\ \dot{n} &= \alpha_n(u)(1 - n) - \beta_n(u)n \\ \dot{h} &= \alpha_h(u)(1 - h) - \beta_h(u)h \end{aligned}$$

Example: Hodgkin-Huxley Model



Model Reductions

The computational complexity and analytical intractability of the Hodgkin-Huxley model have motivated the derivation of **model reductions**, which are based on two-dimensional differential equations (instead of the four-dimensional equation of the Hodgkin-Huxley model) that can be analyzed in the **phase plane**.

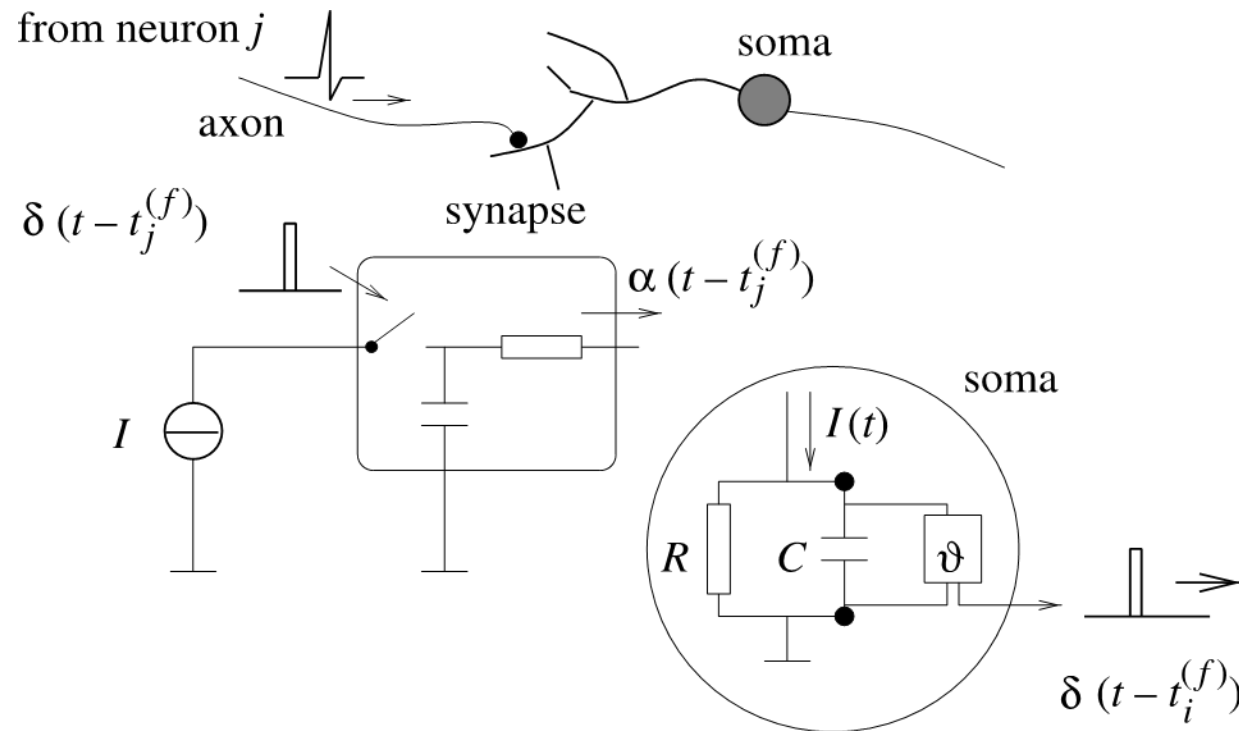
Examples

- Morris-Lecar model
- FitzHugh-Nagumo model

For many purposes, even reduced conductance-based neuron models are more detailed than required and therefore replaced by **phenomenological models** that capture the overall behavior of biological neurons without explicitly emulating the underlying dynamics.

The Leaky Integrate-and-Fire Model

The leaky integrate-and-fire model is a phenomenological neuron model which is based on a simplified electrical circuit with a capacitor C and a resistor R :



From W. Gerstner and W. M. Kistler, Spiking neuron models: Single neurons, populations, plasticity. Cambridge U.K., New York: Cambridge University Press, 2002.

The Leaky Integrate-and-Fire Model

The input current $I(t)$ is split into a component I_R for the resistor and a component I_{cap} for the capacitor:

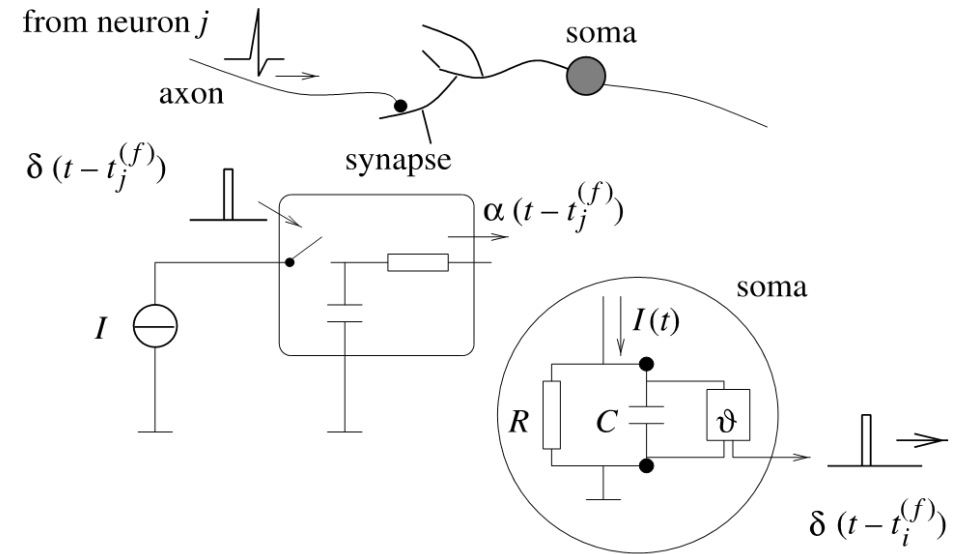
$$I(t) = I_R(t) + I_{cap}(t) = \frac{u(t)}{R} + C \frac{du}{dt}$$

Substituting $\tau_m = RC$, the neural time constant, yields:

$$\tau_m \frac{du}{dt} = -u(t) + RI(t)$$

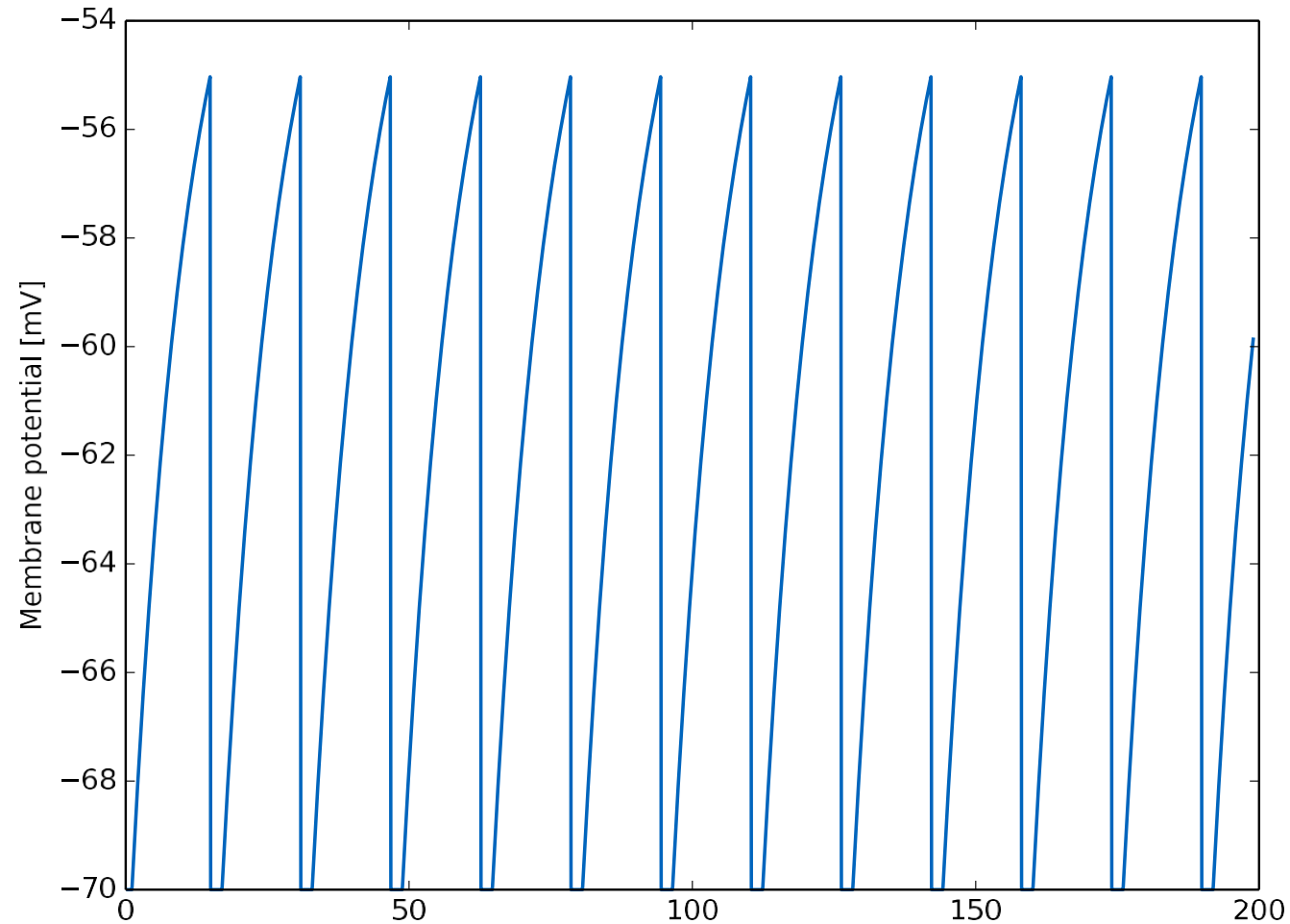
Unlike in the Hodgkin-Huxley model, spike firing times t^f must be detected “manually” based on a threshold ϑ :

$$t^f: u(t^f) = \vartheta$$



From W. Gerstner and W. M. Kistler, Spiking neuron models: Single neurons, populations, plasticity. Cambridge U.K., New York: Cambridge University Press, 2002.

Example: Leaky Integrate-and-Fire Model



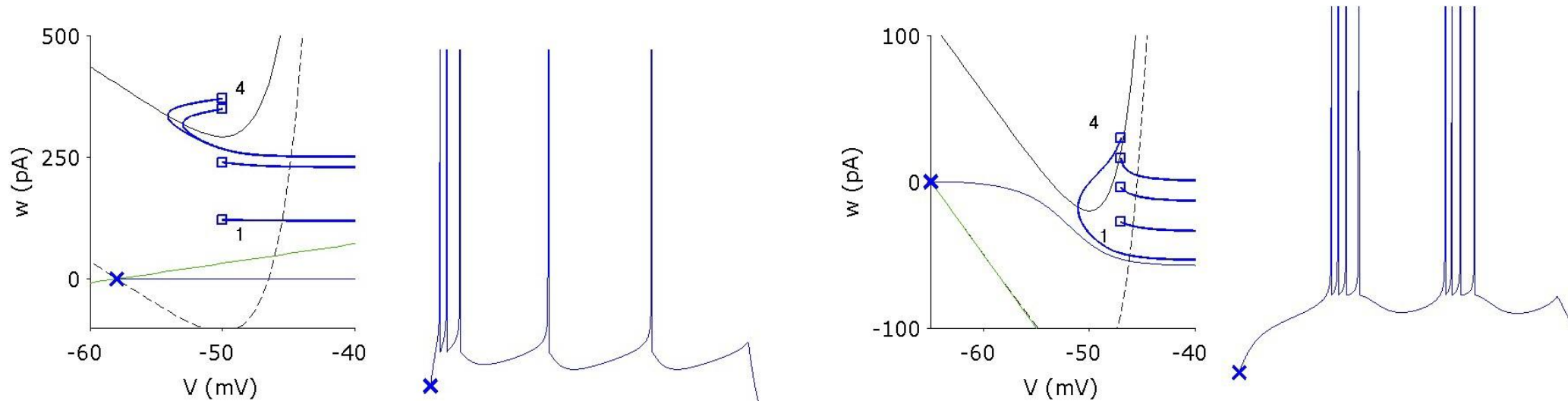
The AdEx Model

The adaptive exponential integrate-and-fire (AdEx) model is an extended **phenomenological** spiking neuron model that combines the capability of the Hodgkin-Huxley model to produce **biologically realistic firing patterns** with the low computational complexity of other phenomenological models. It is based on a two-dimensional differential equation that can be analyzed in the phase space:

$$C \frac{dV}{dt} = -g_L(V - E_L) + g_L \Delta_T \exp\left(\frac{V - V_T}{\Delta_T}\right) - w + I$$

$$\tau_w \frac{dw}{dt} = a(V - E_L) - w$$

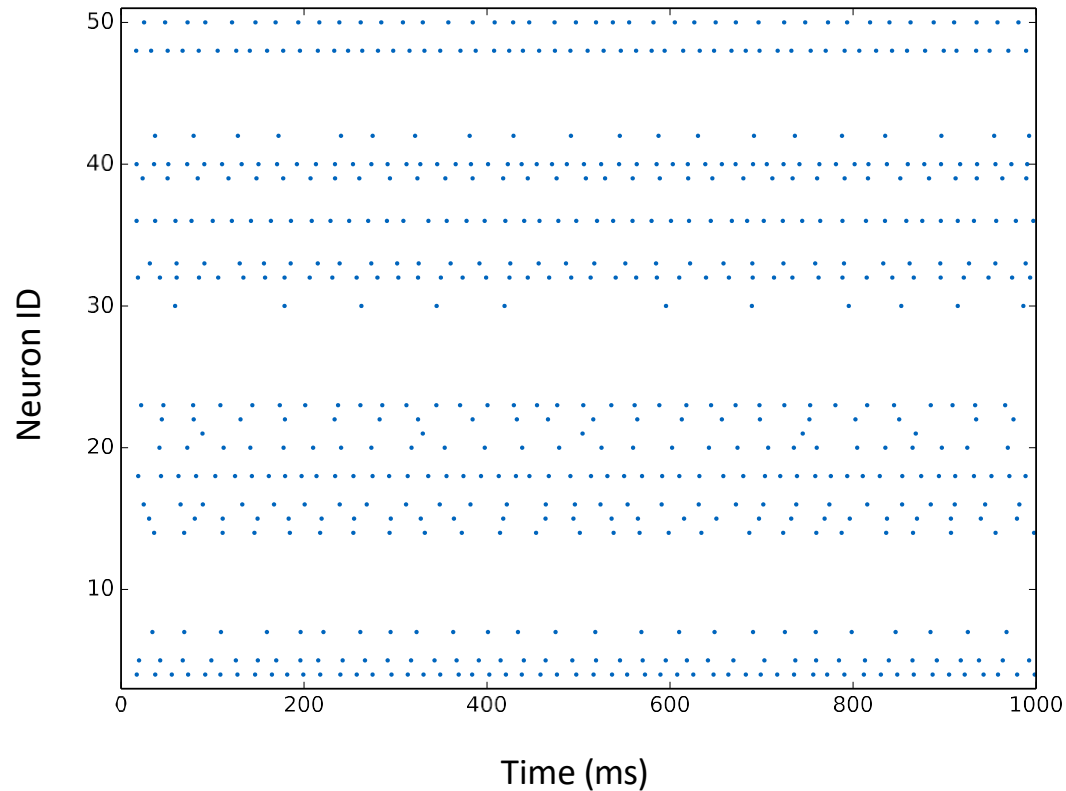
Example: The AdEx Model



From Wulfram Gerstner and Romain Brette (2009) Adaptive exponential integrate-and-fire model. Scholarpedia, 4(6):8427.

Since the AdEx model is based on a two-dimensional differential equation, it can be analyzed in the phase space.

The Challenge of Neural Coding



What is the meaning of this spike train?

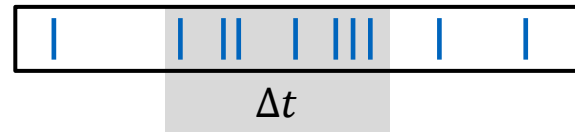
Spike train generated based on http://nest-simulator.org/py_sample/random-balanced-network-exp-synapses-multiple-time-constants/

Neural Codes

A central question in computational neuroscience is how information can be **encoded** in spike trains.

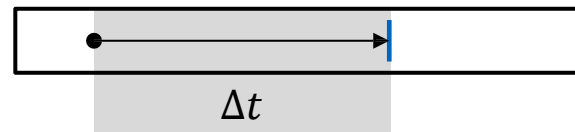
Rate Coding

The information is carried in the averaged spike rate within a certain time window (e.g., muscle excitation).



Time-to-First Spike Coding

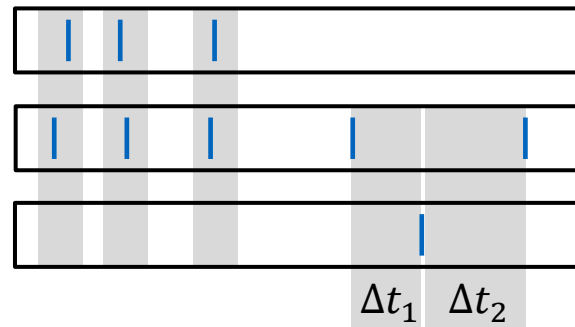
The information is encoded in the time of emission of the first spike after a reference point (e.g., the onset of a stimulus).



Neural Codes

Coding by Synchrony or Correlation

Population codes are based on the activity patterns of a group of neurons. Similarly to single neurons, information can be encoded in the averaged overall activity of the population. Coding by synchrony or correlation is based on the direct or indirect temporal coherence between spikes of individual neurons in a population.



In general, rate codes can be approximated using analog neuron models.

Population Models

In **large populations** of neurons, the contribution of a single neuron becomes more and more **negligible**. Instead of modeling and describing every neuron individually, population models therefore only model the mean activity of a population.

- Populations are modeled based on **density functions** that compute the **probability** that an arbitrary neuron has a certain internal state.
- **Neural field models** do not assume a discrete set of populations but rather a **continuum of neurons** in space.

Spiking Neurons vs. Analog Neurons

Property	Spiking Neurons	Analog Neurons
Model	Dynamical system	Activation function
Communication	“Digital” spikes	“Analog” values
Encoding of Information	Spike timing, rate, and synchrony	Real numbers
Computational Complexity	High – depends on the model	Low
Biological Plausibility	Usually high	Usually low